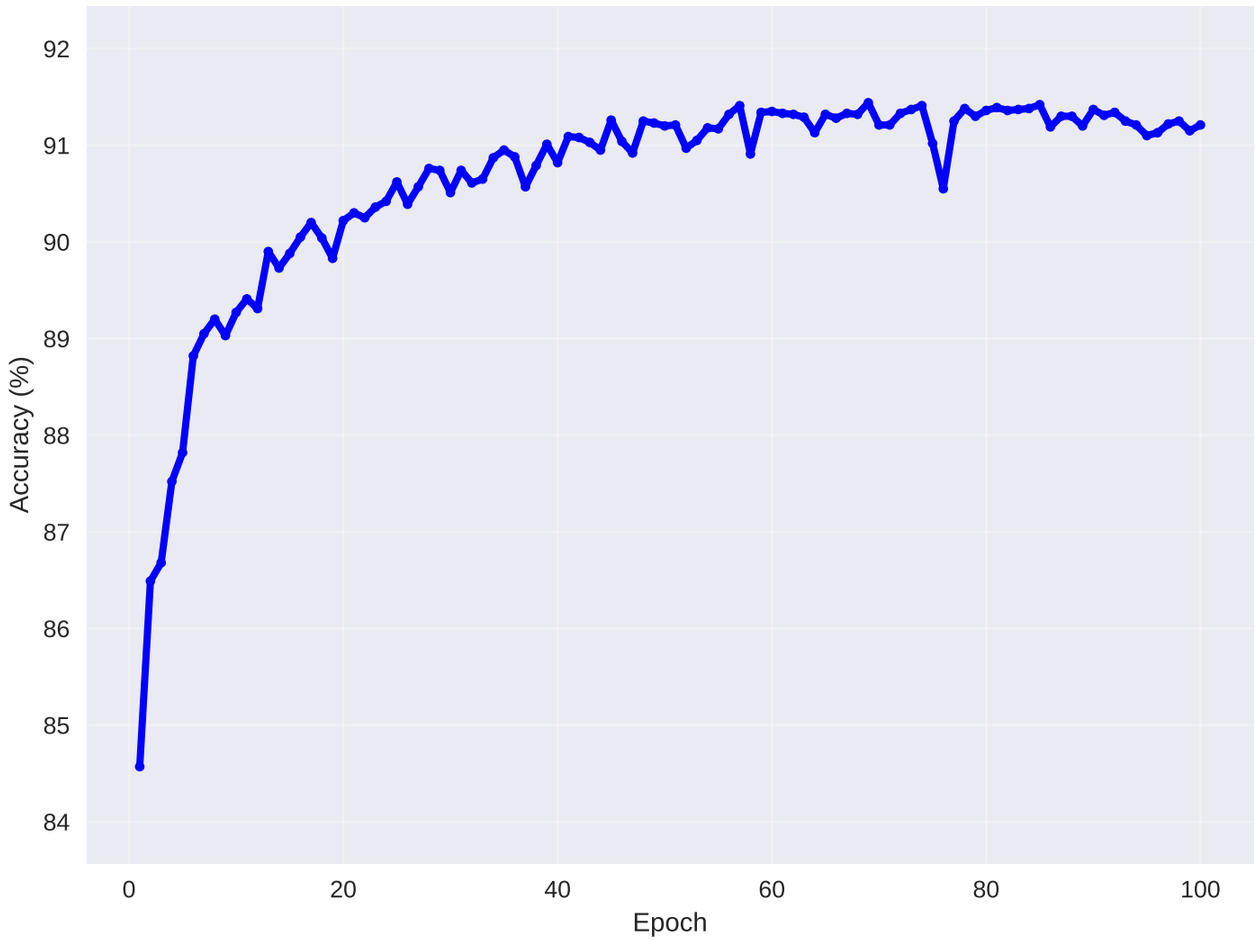
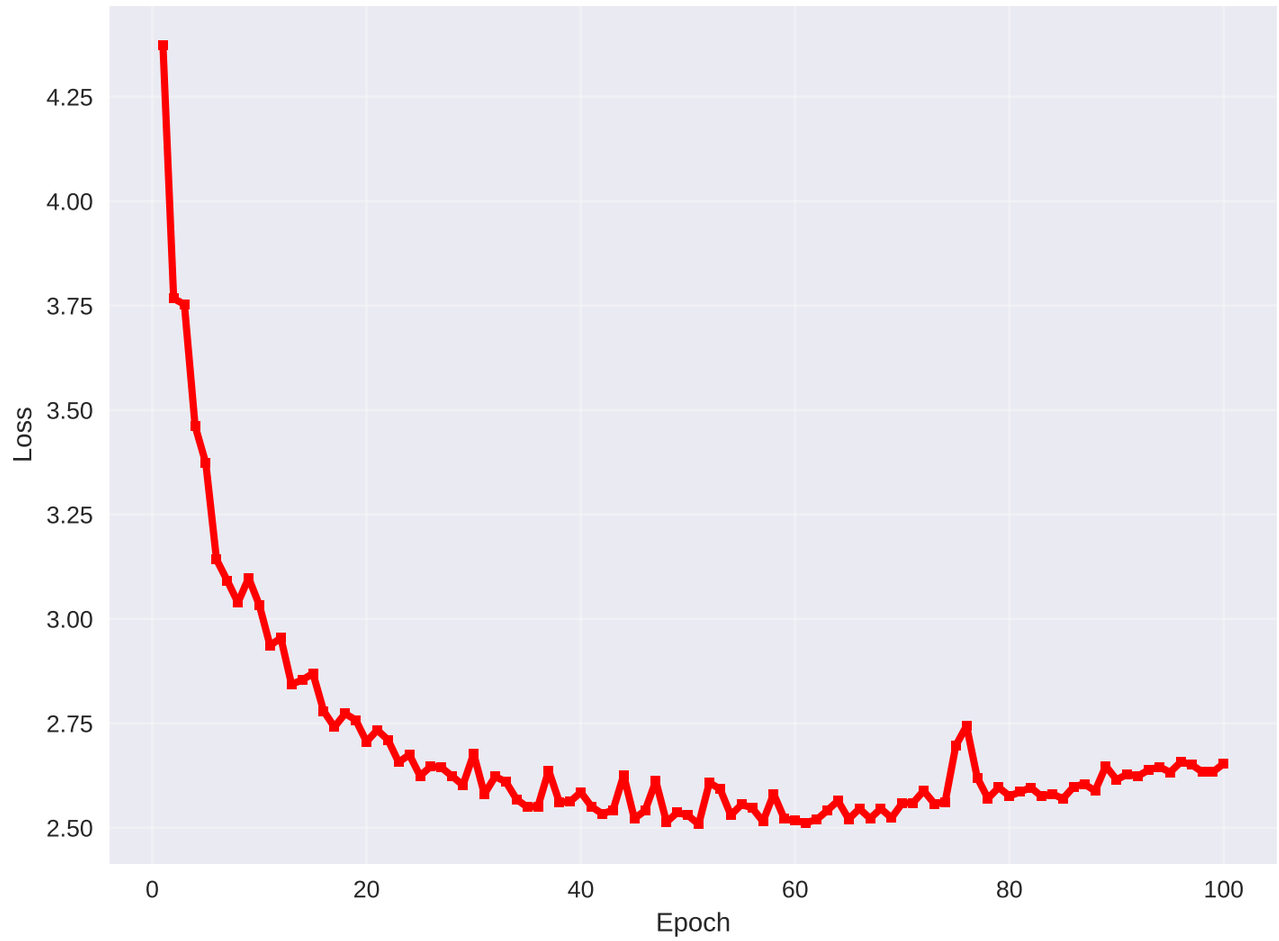
Federated Learning - Main Metrics Overview

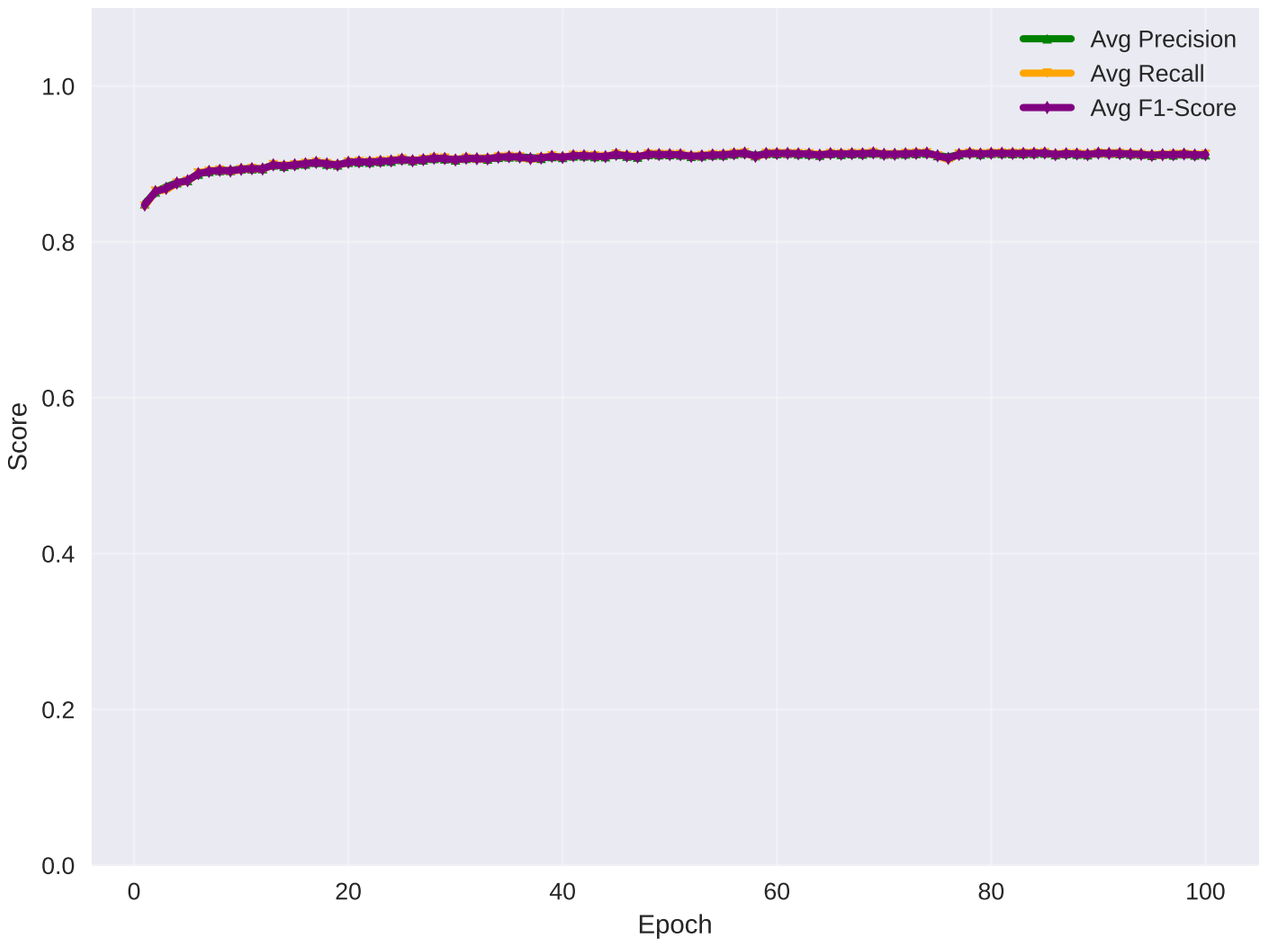
 Global Model Accuracy Evolution



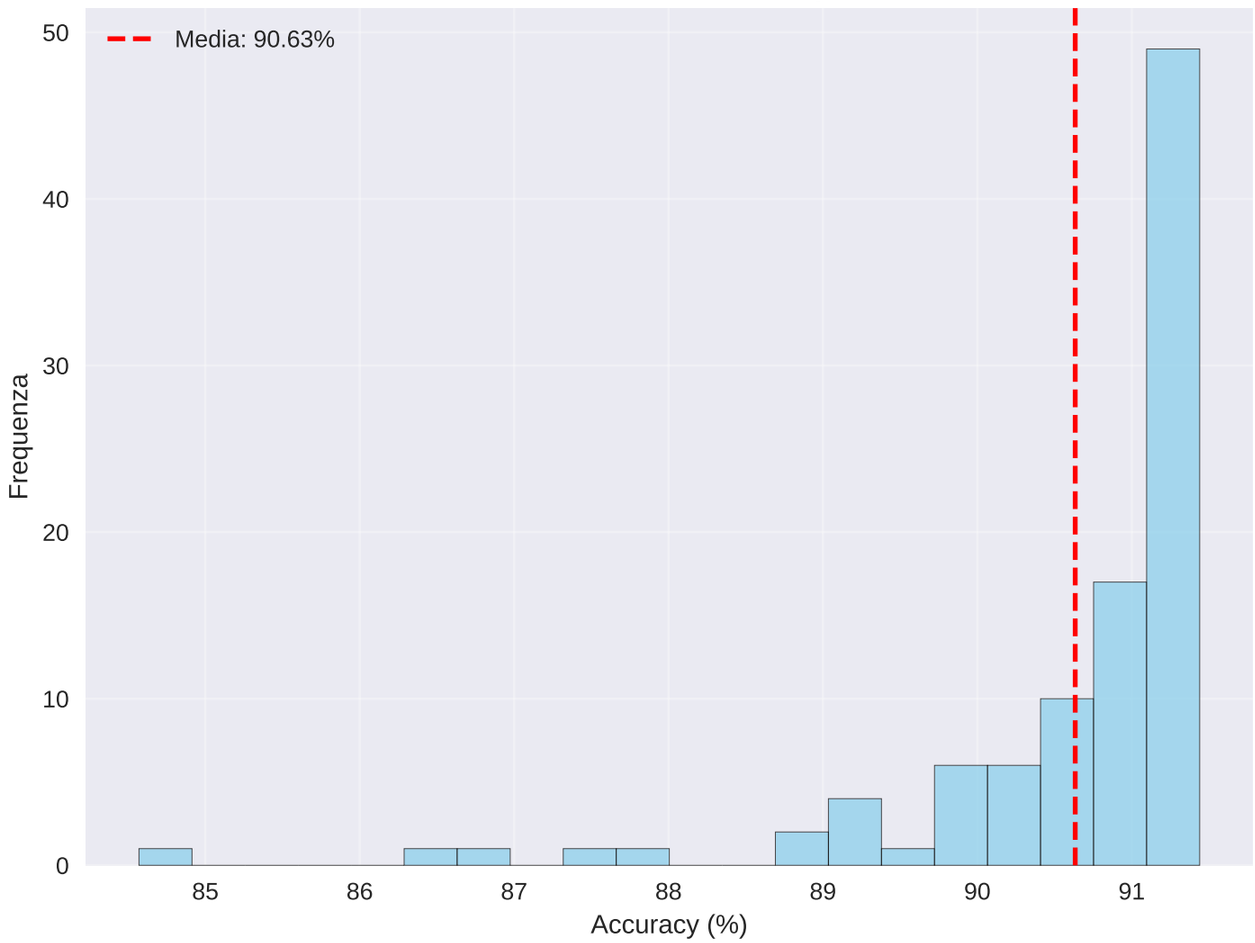
 Global Model Loss Evolution



 Average Precision, Recall & F1-Score

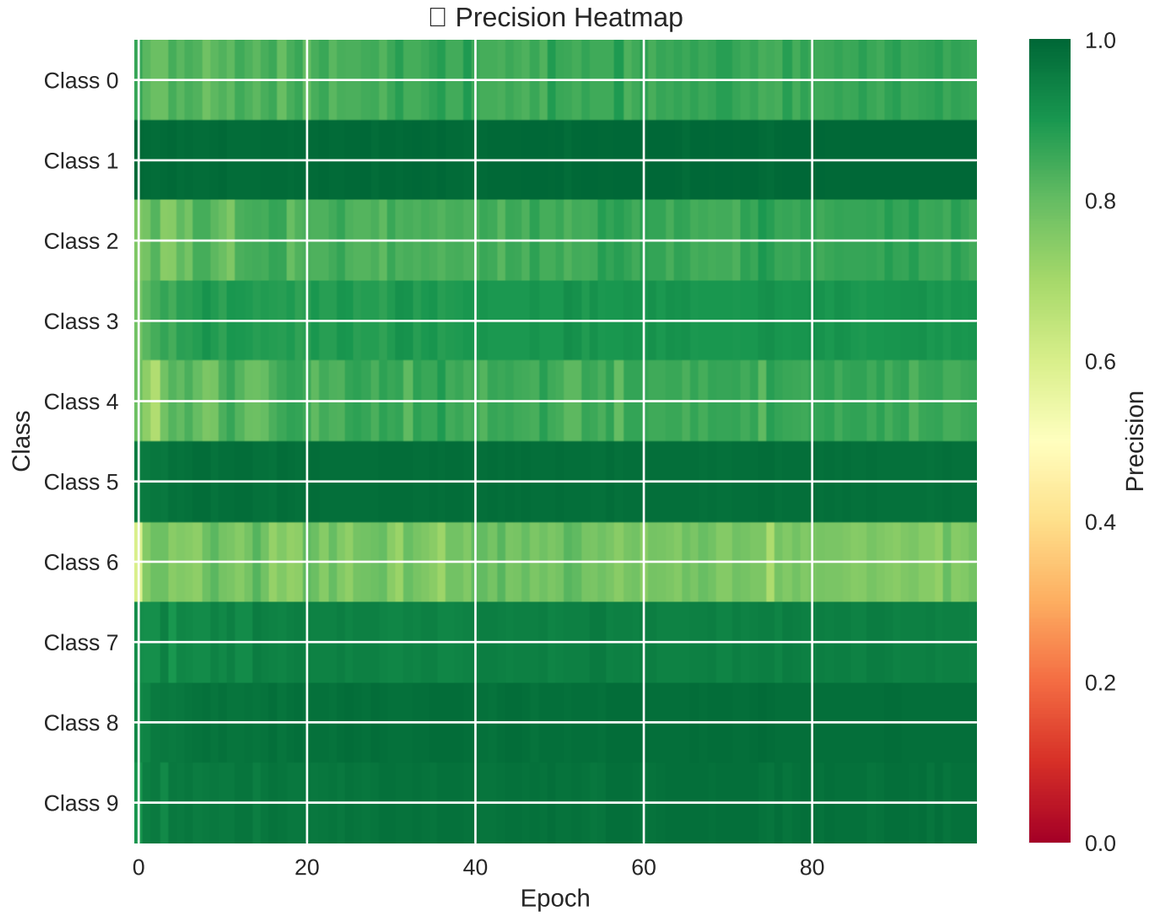


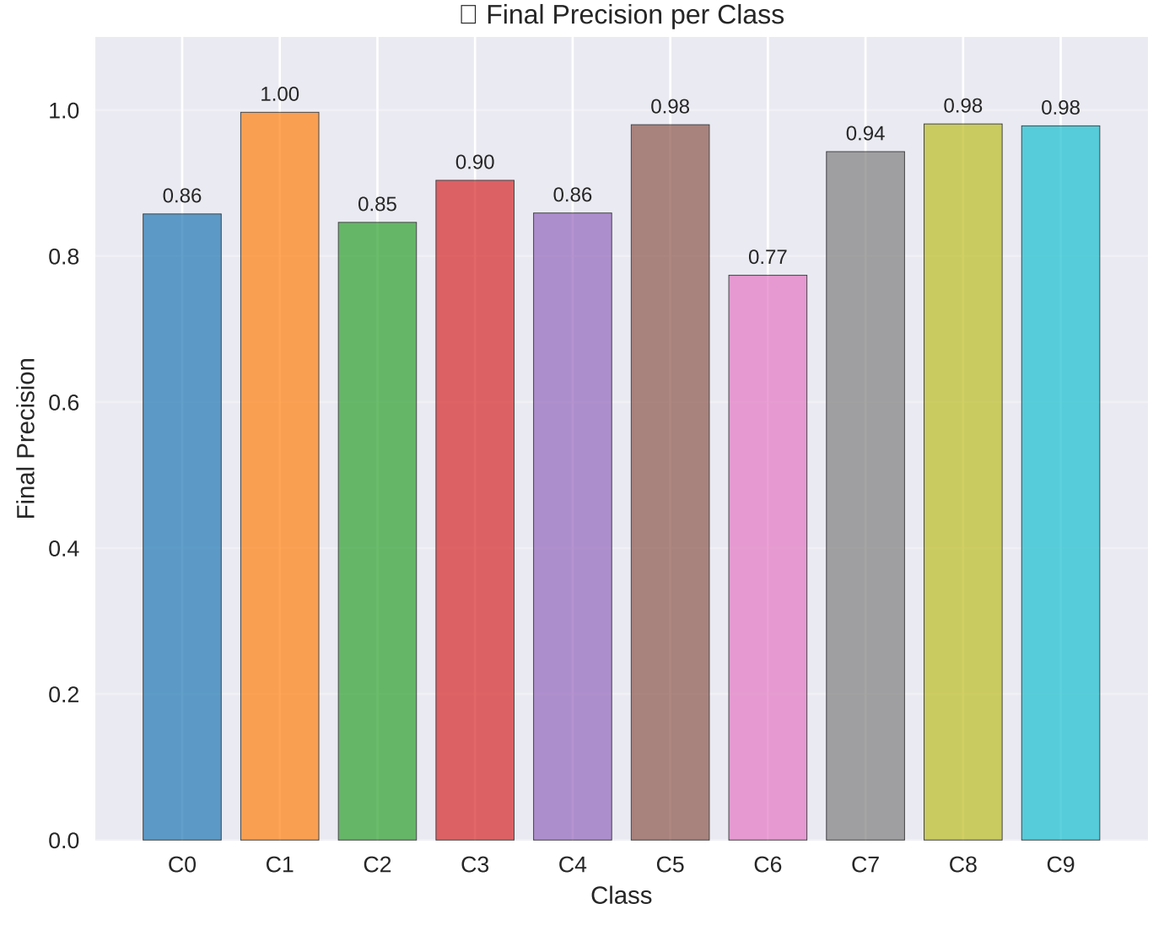
 Distribuzione Accuracy

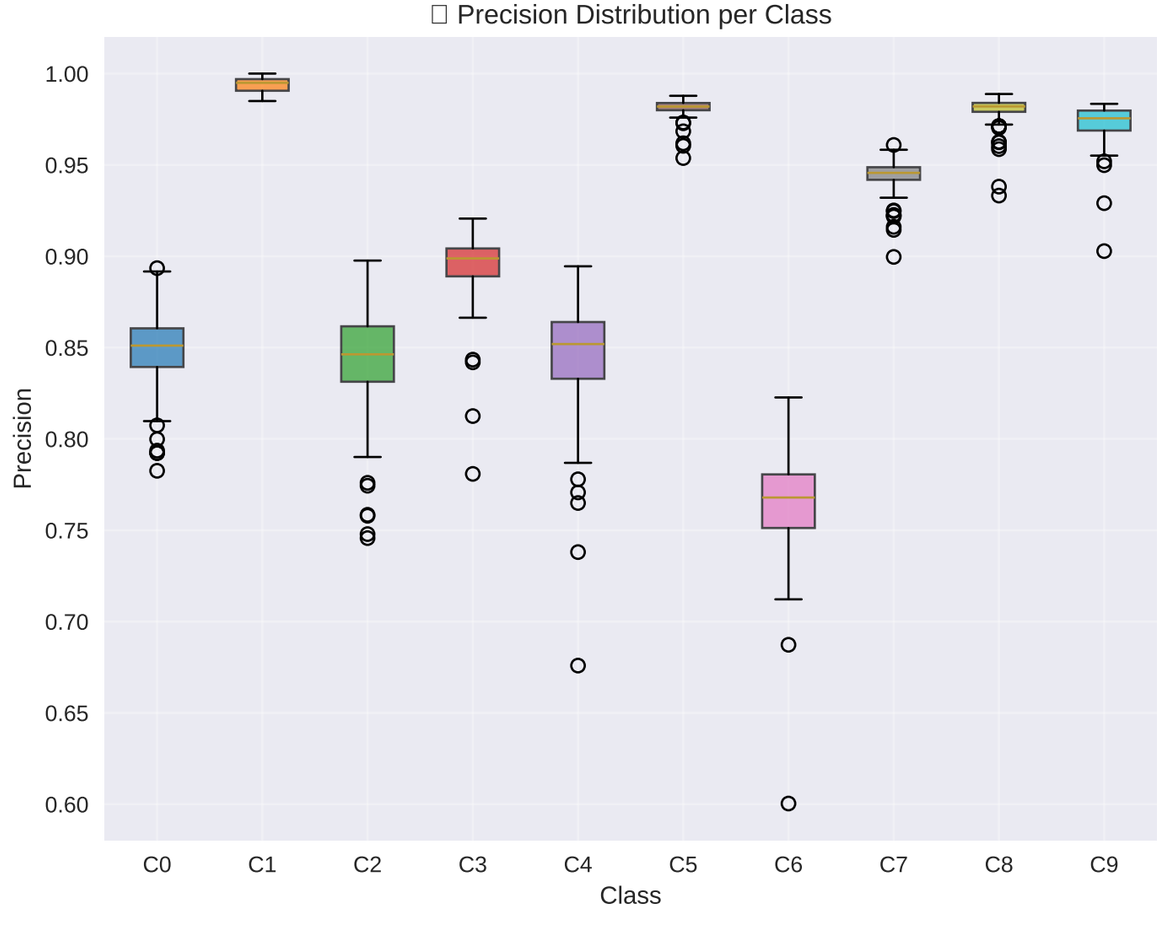


Per-Class Precision Analysis



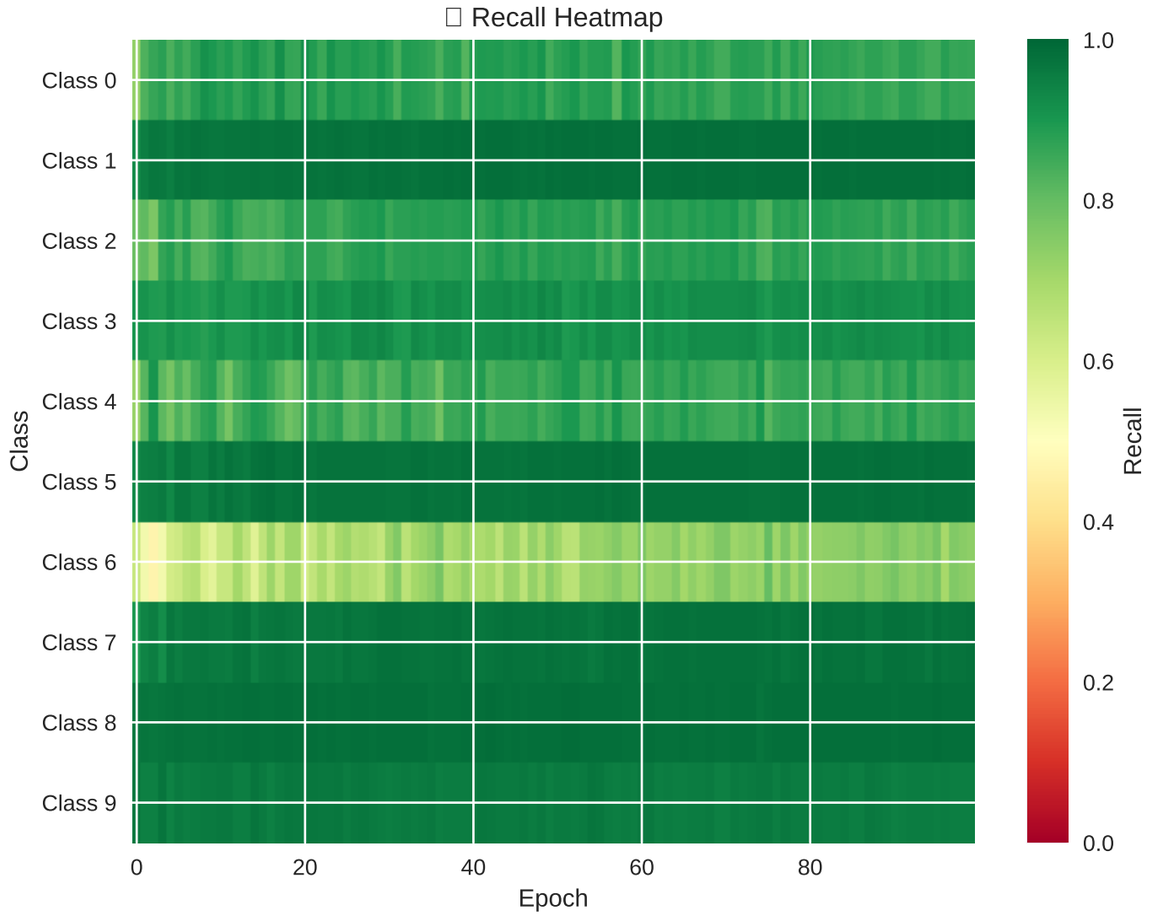


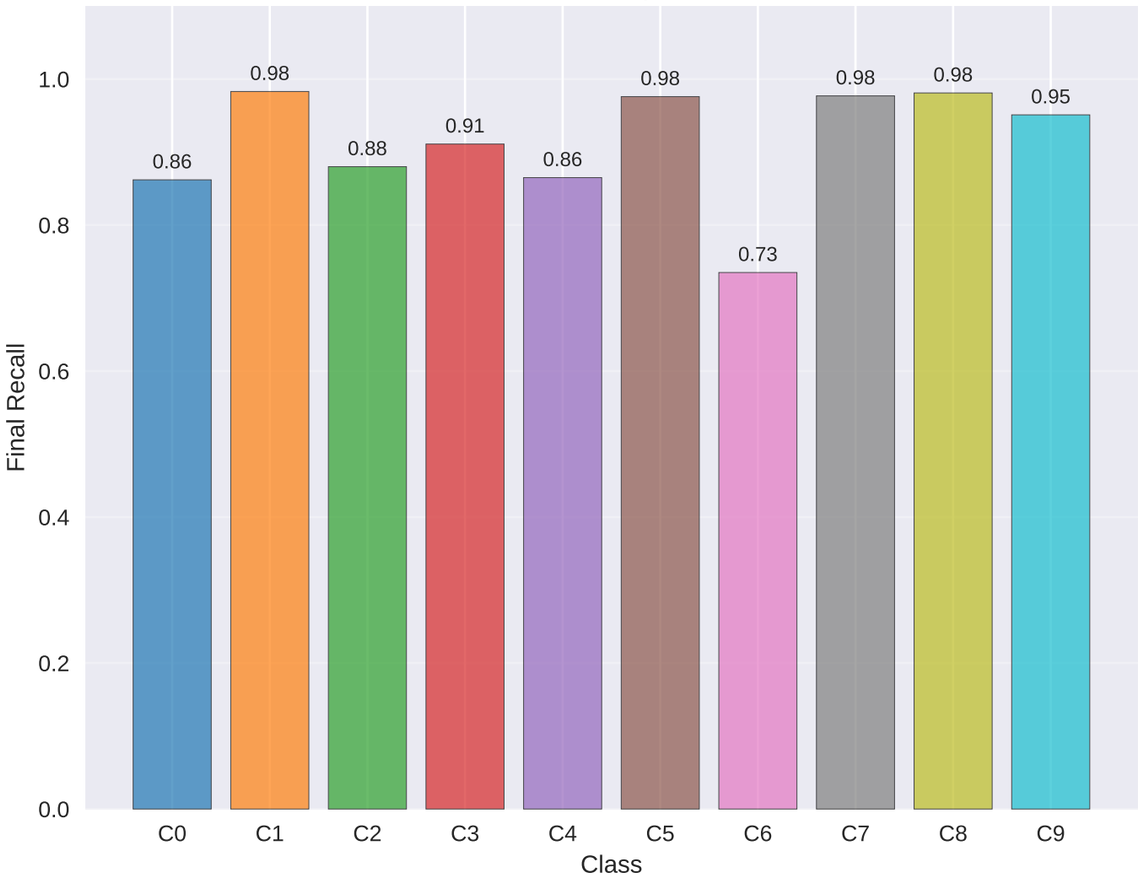


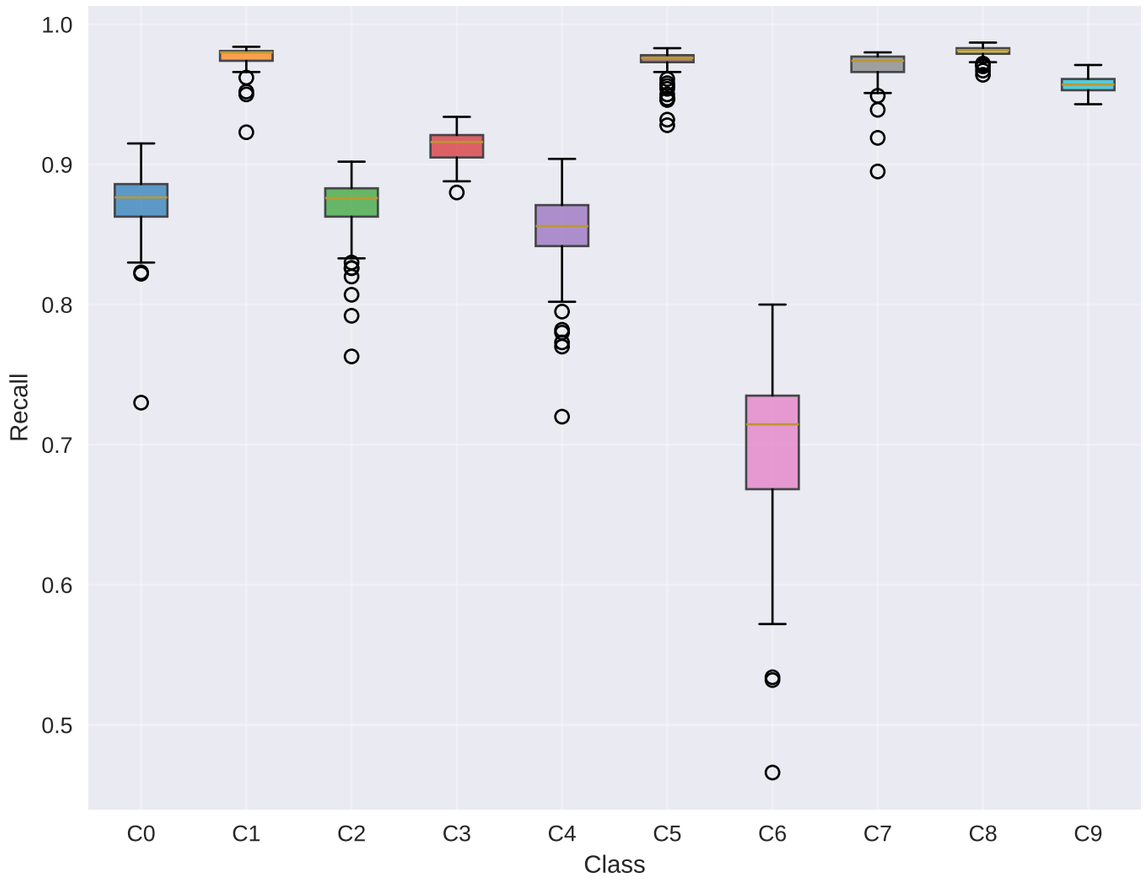


Per-Class Recall Analysis

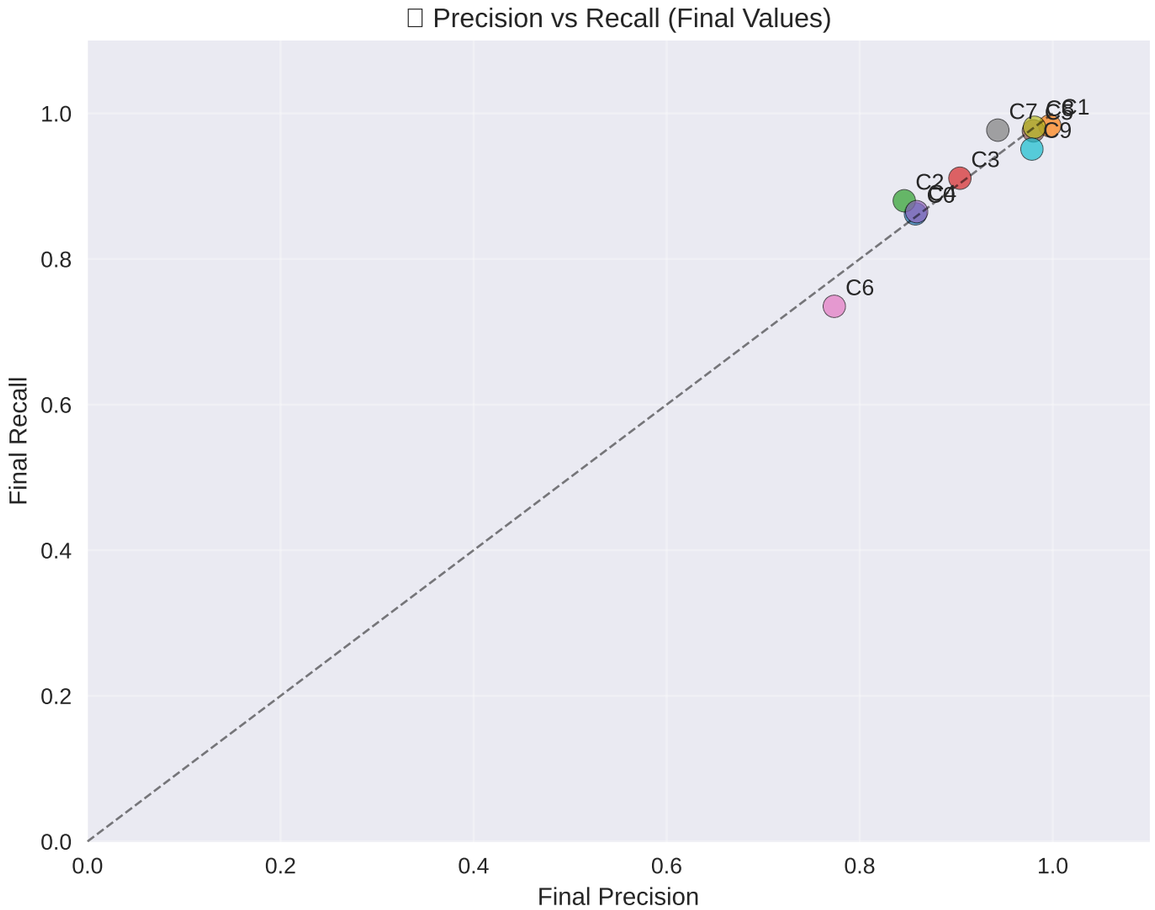




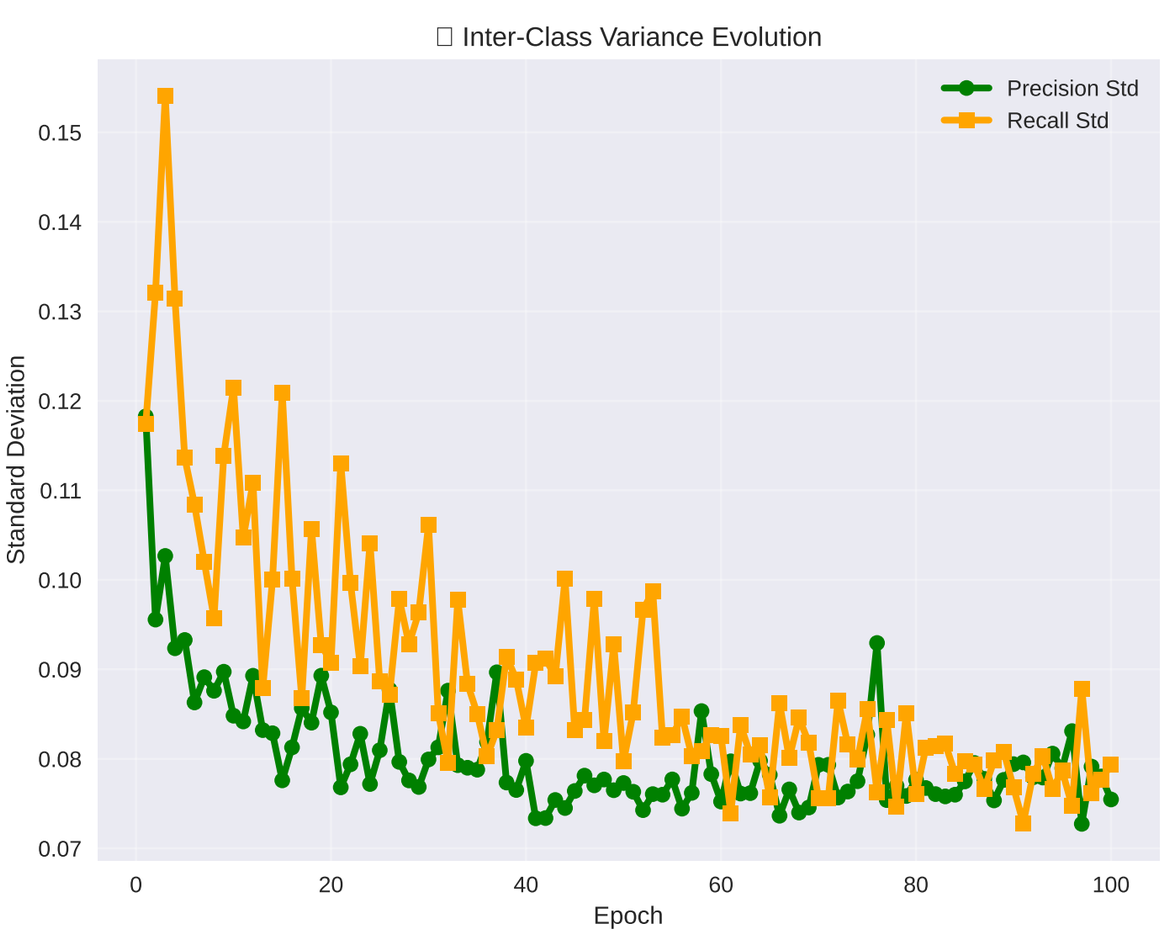


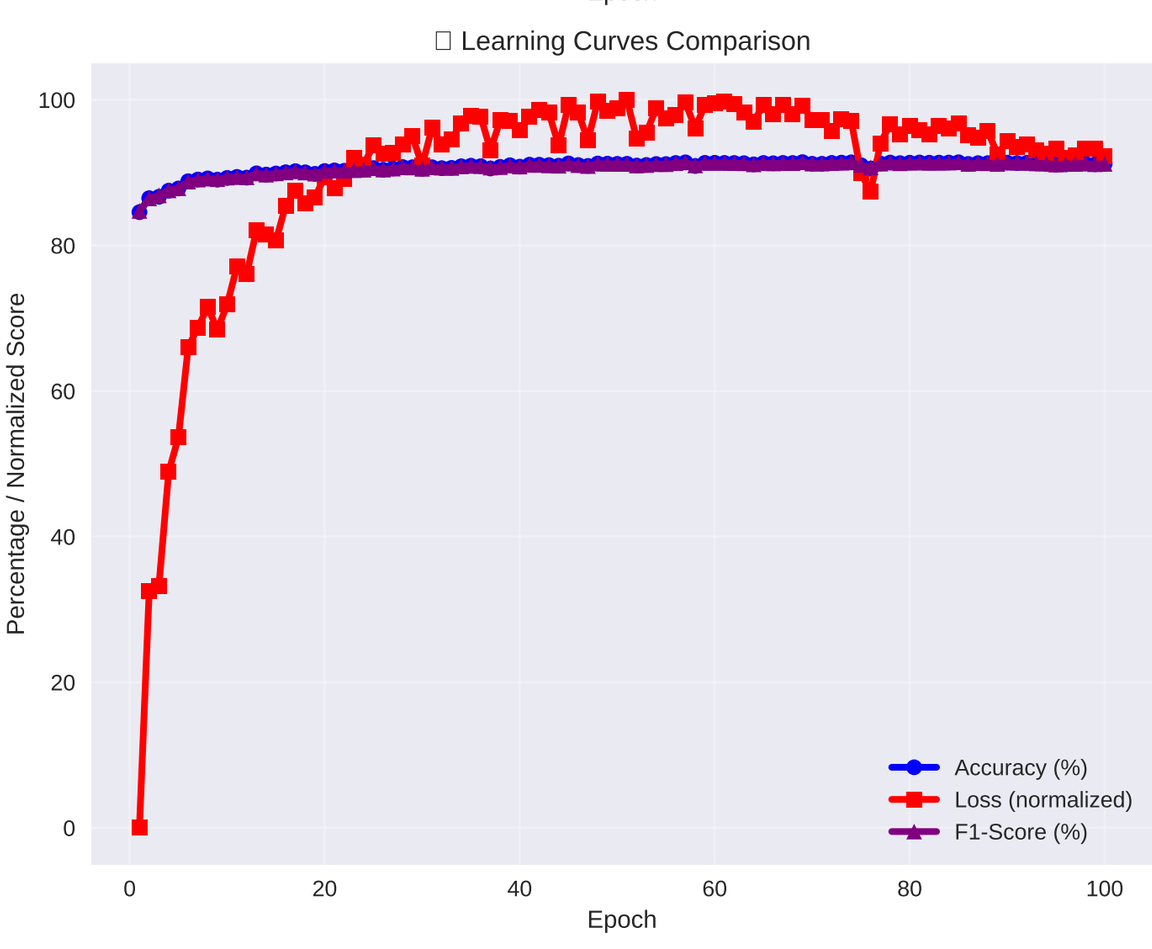


Comparative Analysis









Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-08-29 23:17:14
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 91.21%
• Loss: 2.6534
• Average Precision: 0.912
• Average Recall: 0.912
• Average F1-Score: 0.912

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +6.64%
• Riduzione Loss: +1.7204
• Stabilità Accuracy (CV): 0.0126

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (0.997)
• Peggiore Precision: Classe 6 (0.774)
• Migliore Recall: Classe 1 (0.983)
• Peggiore Recall: Classe 6 (0.735)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.858, R=0.862, F1=0.860
• Classe 1: P=0.997, R=0.983, F1=0.990
• Classe 2: P=0.846, R=0.880, F1=0.863
• Classe 3: P=0.904, R=0.911, F1=0.907
• Classe 4: P=0.859, R=0.865, F1=0.862
• Classe 5: P=0.980, R=0.976, F1=0.978
• Classe 6: P=0.774, R=0.735, F1=0.754
• Classe 7: P=0.943, R=0.977, F1=0.960
• Classe 8: P=0.981, R=0.981, F1=0.981
• Classe 9: P=0.978, R=0.951, F1=0.965

VALUTAZIONE QUALITATIVA:
• Performance Globale: BUONA
• Stabilità Training: STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
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